Benchmarking Minimax Search, Imitation of Human Play, Self-Play Reinforcement Learning, and Monte Carlo Tree Search over Policy-Value Networks to Identify the Strongest Paradigm in a Multi-Agent Xiangqi (象棋) League

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***Abstract*—Modern game-playing agents are built from three ideas validated on Go and chess at cluster scale: imitation of human experts, reinforcement learning by self-play, and tree search over a learned network. It is unknown which of these paradigms is worth its cost for *Xiangqi* (象棋, Chinese Chess) on the single consumer GPU most researchers and students actually use, whether imitation pretraining still helps the subsequent reinforcement learning under that budget, and what factor ultimately bounds playing strength. Existing Xiangqi deep-learning work consists of a few single-system papers and provides no controlled, like-for-like comparison. We propose a controlled, equal-compute study that places four paradigms—classical alpha–beta search, pure-self-play Proximal Policy Optimization, imitation-initialised Proximal Policy Optimization, and AlphaZero-style Monte Carlo Tree Search—on a single shared Xiangqi engine, a fixed 2086-move action space, one policy–value network, and one Elo scale, so the paradigm is the only variable. We rank the agents by fitted Elo with 95% confidence intervals, quantify the effect of imitation pretraining on the subsequent reinforcement learning, and run a four-lever ablation (search, epochs, teacher, network size) to identify the binding constraint. A complete, tested implementation and preliminary runs over a 9.3M-position training split from 141k human games already exist, making the study a rigorous formalisation rather than a speculative build. The expected contributions are a reusable multi-agent evaluation framework, to our knowledge the first controlled paradigm comparison for Xiangqi under constraint, and reproducibility guidance for self-play reinforcement learning.**

***Index Terms*—Xiangqi, Chinese Chess, deep reinforcement learning, Monte Carlo tree search, imitation learning, Proximal Policy Optimization, Elo rating, empirical software engineering, reproducibility.**

I. Introduction

THE dominant recipe for superhuman board-game play was established on Go by AlphaGo, which combined supervised (imitation) learning of expert moves, reinforcement learning by self-play, and Monte Carlo Tree Search [1]. AlphaGo Zero removed the human-data stage and learned *tabula rasa* [2], and AlphaZero generalised the same algorithm to chess, shogi, and Go, replacing alpha–beta search with general Monte Carlo Tree Search [3]. These landmark results share two properties that limit their guidance for most practitioners: they were obtained on large tensor-processing clusters, and each reports a single successful pipeline rather than a controlled comparison of the paradigms it combines.

Motivation and problem. For a researcher, student, or practitioner working on a *single consumer GPU*, the practical questions are unanswered, especially for *Xiangqi* (象棋, Chinese Chess)—a culturally major but academically under-served game with a larger 10×9 board, higher branching than chess, a cannon that captures by jumping a screen, and the unusual rule that *stalemate is a loss*. The published deep-learning work on Xiangqi is recent and sparse: an AlphaZero-style player that searches with Monte Carlo Tree Search [4], a population method that tackles strategic non-transitivity [5], and a language-model approach trained with policy optimization [6]. Each builds *one* maximally strong system; none measures, under a fixed budget, how the distinct learning paradigms compare, whether imitation pretraining helps the reinforcement learning that follows, or which factor limits strength.

Significance. Deep reinforcement-learning results are known to be fragile and hard to reproduce when reported from single runs without variance analysis [7]. A controlled, confidence-interval-backed comparison on one rule engine therefore yields decision-relevant, reproducible evidence that cluster-scale, single-system papers do not.

Research questions. We pose three questions, each answerable by the methodology below:

- **RQ1 (ranking):** under a fixed single-GPU budget, how do the four paradigms rank in Xiangqi playing strength on a common Elo scale?
- **RQ2 (effect of imitation pretraining):** does imitation pretraining improve the subsequent Proximal Policy Optimization self-play under that budget, or can reinforcement learning fine-tuning *degrade* the imitation policy?
- **RQ3 (binding constraint):** which factor—play-time search, model capacity, training data or teacher, or the learning algorithm—most limits playing strength at this

scale?

Objectives and contributions. We will (i) rank the four paradigms on a common Xiangqi Elo scale under matched compute; (ii) quantify the signed effect of imitation pretraining on the subsequent Proximal Policy Optimization; and (iii) identify, through a single-variable ablation, whether search, model capacity, training data, or the learning algorithm binds strength. The contributions are: a reusable single-interface *multi-agent evaluation league* for heterogeneous game-playing paradigms runnable on one GPU; a controlled, equal-budget, Elo-anchored paradigm comparison for Xiangqi, including negative results; and concrete reproducibility guidance for self-play reinforcement learning.

II. Related Work

Deep learning with search. The AlphaGo family established the two levers we compare—imitation pretraining and search over a learned network [1]–[3]; Monte Carlo Tree Search itself is surveyed comprehensively by Browne *et al.* [8], whose selection rules our search agent builds on. Classical Xiangqi engines instead use alpha–beta search with handcrafted evaluation, the tradition our Minimax baseline represents.

Policy gradients and reproducibility. Proximal Policy Optimization [9] is our reinforcement-learning optimizer, chosen for stability; yet Henderson *et al.* [7] show policy-gradient results vary sharply with seeds and implementation, motivating our interval-based evaluation and a training-stability analysis.

Imitation and its ceiling. The AlphaGo pipeline initialised its policy by imitating human experts before refining it with reinforcement learning [1]; a known limitation is that an imitator cannot surpass its demonstrators, the usual justification for the reinforcement-learning stage. Whether that stage helps *under constrained compute*, however, is untested for Xiangqi—and our preliminary data suggest it can hurt, which is precisely what a controlled study should resolve.

Populations and evaluation. AlphaStar extended self-play to a league of adapting agents rated on an Elo ladder [10]; we adopt that framing for a league of *heterogeneous* agents. For Xiangqi, JiangJun combines Monte Carlo Tree Search with Policy-Space Response Oracles to address non-transitivity, analysing >10,000 human games [5]—a reminder that single head-to-head results can mislead and that round-robins with fitted Elo are needed.

Xiangqi-specific deep learning. The directly comparable systems [4]–[6] each optimise one pipeline at either cluster scale or without cross-paradigm comparison. Table I summarises the resulting gap.

Research gap. No controlled, equal-compute, Elo-anchored comparison of classical search, pure Proximal Policy Optimization, imitation-initialised Proximal Policy Optimization, and AlphaZero-style Monte Carlo Tree Search exists for Xiangqi under a single-GPU budget; consequently the direction of the imitation-pretraining effect and the binding constraint on strength are both unquantified for this game at this scale. This is primarily an *evaluation/methodological* gap, with secondary *performance* and *application* dimensions.

TABLE I
Where prior work leaves the gap this study fills.

Work	Xiangqi	Cross-paradigm comparison	Constrained GPU budget
AlphaGo family [1]–[3]	no	no	no
Monte Carlo Tree Search survey [8]	n/a	n/a	n/a
Proximal Policy Optimization [9]	no	partial	no
Reproducibility [7]	no	no	yes
AlphaStar [10]	no	no	no
JiangJun [5]	yes	no	no
Yilmaz *et al.* [4]	yes	no	partial
Xiangqi-R1 [6]	yes	no	partial
This study	**yes**	**yes**	**yes**

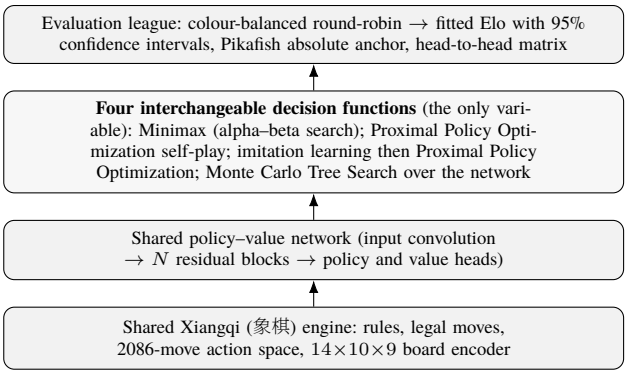


Fig. 1. The controlled apparatus. One environment, one action space, and one network are shared by all four agents; only the decision function changes, and all agents are rated on one Elo scale. (Arrows denote data/control flow.)

III. Proposed Methodology

Design. A controlled comparative study: the independent variable is the paradigm (or the one ablated lever); the dependent variable is Elo; everything else is fixed by a shared stack (Fig. 1), so differences in strength are attributable to the decision function alone. "Equal effort" is operationalised as a matched number of self-play environment steps per reinforcement-learning agent on one RTX-4060-class GPU, reported beside every result so the comparison is budget-explicit rather than open-ended.

Shared components. (1) A tested Xiangqi rule engine exposed through a Gymnasium interface, with a fixed 2086-move action space and legal masking, and a $14 \times 10 \times 9$ side-to-move-relative board encoding. (2) One AlphaZero-style policy–value network reused by imitation, by Proximal Policy Optimization (as actor and critic), and by Monte Carlo Tree Search (as priors and leaf value); its width and depth are configurable for the ablation.

The four agents. *Minimax*: alpha–beta (negamax) search with a material evaluation. *From-scratch Proximal Policy Optimization*: action-masked self-play from random initialisation. *Imitation-initialised Proximal Policy Optimization*: supervised imitation of 113k human games, then action-masked self-play refinement from that checkpoint. *Monte Carlo Tree Search agent*: a best-first search that uses the network's priors to guide selection and its value head to evaluate leaves, choosing moves by visit count. A random agent is included as a floor.

TABLE II
TRACEABILITY: EACH RESEARCH QUESTION MAPS TO AN OBJECTIVE, AN EXPERIMENT, AND A PRIMARY METRIC.

Question	Objective	Experiment	Primary metric
RQ1	Rank the paradigms	Experiment 1	fitted Elo + 95% CI
RQ2	Imitation-pretraining effect	Experiment 2	signed Elo difference
RQ3	Binding constraint	Experiment 3	per-lever Elo effect

Data. 141,356 legal human games (≈11.6M positions) are split at the game level into 113k/14k/14k train/validation/test sets, recorded in a committed manifest for reproducibility; an engine self-play corpus supports the teacher-distillation lever.

Procedure. Each reinforcement-learning paradigm is trained to a matched single-GPU budget with periodic checkpointing. *Experiment 1 (RQ1)* runs a colour-balanced round-robin among all four agents plus the random agent and fits Elo with 95% confidence intervals using (1) and a head-to-head matrix. *Experiment 2 (RQ2)* isolates imitation-only, imitation-initialised Proximal Policy Optimization, and from-scratch Proximal Policy Optimization and reports the signed Elo differences plus learning proxies (imitation top-1 accuracy; self-play reward and value traces). *Experiment 3 (RQ3)* varies one lever at a time—search simulations $\in \{100, 200, 400, 800\}$, imitation epochs, teacher (human versus engine distillation), and network size (1.15M versus 4.85M parameters)—each re-entered into the round-robin, reporting each lever's Elo effect. A stability study compares fragmented, continuous, and replay-buffer-persisted training.

Evaluation metrics. The expected score of a rating difference is

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}, \tag{1}$$

inverted and fitted over round-robin results to obtain each agent's rating; confidence intervals follow the delta method. Reported metrics are fitted and absolute Elo (with 95% confidence interval), head-to-head win rates, the signed Elo difference between the imitation-only and imitation-initialised agents, per-lever Elo effects, top-1 move accuracy, value-loss trajectories, and GPU-hour cost. Differences are called significant only when confidence intervals do not overlap; non-transitivity is handled by full round-robins and fitted Elo rather than single matches [5]. Fixed seeds, the committed manifest, and per-run configuration, seed, and commit logging address known reproducibility pitfalls [7].

Threats to validity. Relative Elo depends on the agent pool; we mitigate this with an absolute Pikafish anchor and full round-robins rather than single matches. Few games per pair widen confidence intervals, so the decisive comparisons will be rerun with more games until their intervals separate. Findings are scoped to one game, one network family, and one GPU class, and we claim no automatic generalisation beyond that setting; the shared-stack design nonetheless keeps the comparison internally valid, since only the decision function varies.

TABLE III
PRELIMINARY FEASIBILITY MEASUREMENTS (RELATIVE ELO, ANCHOR 1500). THESE MOTIVATE THE HYPOTHESES; THE PROPOSED STUDY TIGHTENS AND EXTENDS THEM. A STABILISED RERUN REPRODUCED THE SAME TOP RANK (MONTE CARLO TREE SEARCH ≈1691).

Agent	Elo (prelim.)	95% CI
Monte Carlo Tree Search + network (AlphaZero-style)	1703	±94
Imitation-only	1557	±77
Minimax (d=2)	1541	±77
Imitation then Proximal Policy Optimization	1411	±79
Random	1287	±96

IV. EXPECTED RESULTS AND CONTRIBUTIONS

Expected outcomes. We hypothesise that (H1) the Monte Carlo Tree Search agent attains the highest Elo; (H2) imitation pretraining raises *starting* strength but, under the constrained budget, self-play reinforcement learning does *not* raise the ceiling and may lower it; and (H3) model capacity and training data—not search depth or the learning algorithm—bind strength. Preliminary runs on the existing implementation are consistent with all three and establish feasibility (Table III); these are *preliminary, confidence-interval-wide* measurements that the proposed study will formalise with more games and tighter intervals, not final results.

Contributions. (1) *Technical*: a reusable, single-interface multi-agent evaluation league for comparing heterogeneous game-playing paradigms on one rule engine, runnable on a single consumer GPU. (2) *Research*: a controlled, equal-budget, Elo-anchored comparison of classical search, Proximal Policy Optimization, imitation-initialised Proximal Policy Optimization, and AlphaZero-style Monte Carlo Tree Search for Xiangqi; a quantified answer to whether imitation pretraining helps the subsequent reinforcement learning under constraint; and a multi-lever capacity-bottleneck analysis. (3) *Software engineering*: evidence that training-stability engineering—replay-buffer continuity and best-checkpoint selection—materially affects learned-model quality, yielding actionable reproducibility guidance for self-play reinforcement learning. *Practical implication*: practitioners gain budget-aware guidance on which paradigm is worth its compute, and on whether to invest in search and imitation before reinforcement-learning fine-tuning or in model capacity.

V. CONCLUSION

We have proposed a controlled, Elo-anchored comparison of four deep-learning paradigms for Xiangqi (象棋) under a constrained compute budget, built on a shared environment, action space, and network so that only the decision function varies. The study answers which paradigm is strongest, whether imitation pretraining helps subsequent reinforcement learning, and what factor bounds strength—questions left open by a Xiangqi literature of single, mostly cluster-scale systems. Because a complete, tested implementation and preliminary results already exist, the work is a rigorous, reproducible formalisation whose contributions—a reusable comparison framework, a constrained-budget paradigm comparison for

Xiangqi, and training-stability guidance—are defensible and honestly scoped, including where the evidence is negative.

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